

B.Tech. Degree IV Semester Regular/Supplementary Examination June 2024

CS 19-202-0406 DATABASE MANAGEMENT SYSTEMS (2019 Scheme)

Maximum Marks: 60

Time: 3 Hours

Course Outcome

On successful completion of the course, the students will be able to:

- CO1: Outline the characteristics and features of database systems.
 CO2: Represent the components and relations through an ER diagram and convert that to relational model.
 CO3: Familiarise with the storage structures, accessing methods and indexing techniques.
 CO4: Formulate relational algebra queries according to user requirements.
 CO5: Formulate efficient SQL query and refine it with procedures, cursors etc.
 CO6: Improve the database design by applying normalisation techniques.
 CO7: Familiarise with an SQL interface of a multi-user relational DBMS package to create, secure, populate, maintain and query a database.
 CO8: Discover the basic issues of transactions and concurrency control of them.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate,
 L6 – Create

PO – Programme Outcome

PART A(Answer *ALL* questions)

(8 × 3 = 24)

	Marks	BL	CO	PO
1. (a) Explain the need for database schema. Draw an example schema and compare it with 'Database state'.	3	L2	1	1,2,3,4
(b) Explain the differences among a relationship instance, a relationship type and a relationship set.	3	L2	2	1,2
(c) Discuss the process of disk initialization.	3	L1	3	1,2,4
(d) Differentiate spanned and unspanned records.	3	L2	3	1,2,4
(e) Explain BCNF with help of an example.	3	L2	6	2
(f) Consider the following relations for a database that keeps track of business trips of salespersons in a sales office. SALESPERSON(Ssn, Name, StartYear, DeptNo) TRIP(Ssn, FromCity, ToCity, DepartureDate, ReturnDate, TripId) EXPENSE(TripId, AccountNo, Amount)	3	L2	4	1,3
(i) Write relation algebra expression to get the details of salespersons who have travelled between Mumbai and Delhi and the travel expense is greater than ₹50000.				
(ii) To get the details of salesperson who haven't made any trips.	3	L1	8	1,3
(g) Discuss the problems of deadlock and starvation. Also describe the wait-die and wound-wait protocols for deadlock prevention.	3	L3	8	1,3
(h) Check whether the given schedule S is conflict serializable or not- S: R1(A), R2(A), R1(B), R2(B), R3(B), W1(A), W2(B).				

(P.T.O.)

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		Marks	BL	CO	PO
PART B					
(4 × 12 = 48)					
II.	(a) ✓ Construct an E-R diagram for a car insurance company whose customers own one or more cars each. Each car has associated with it, zero to any number of recorded accidents. Each insurance policy covers one or more cars and has one or more premium payments associated with it. Each payment is for a particular period of time and has an associated due date and the date when the payment was received.	8	L2	2	1,2
	(b) ✓ When is the concept of a weak entity used in data modeling? Define the terms, owner entity type, weak entity type, identifying relationship type and partial key.	4	L1	2	1,2
OR					
III.	(a) Consider University database for the scheduling of classrooms for final exams. This database could be modelled with details as: .course has name, department and c-number .section for each course has s-number and enrollment .room has r-number, capacity and building name .exam has exam-id and time. Illustrate an E-R diagram for this database.	8	L1	2	1
	(b) What is the difference between specialization and generalization? Discuss the two main types of constraints on specializations and generalizations.	4	L2	2	1
IV.	(a) Illustrate structure of B-Tree and B+ Tree and explain how they are different.	6	L2	3	1,3,4
	(b) Discuss any two hashing techniques that allow dynamic file expansion.	6	L1	3	1,3,4
OR					
V.	(a) ✓ What is meant by spanned and unspanned records?	6	L2	3	1,3,4
	(b) ✓ Explain the different index structure for files.	6	L1	3	1,3,4
VI.	(a) For the relation schema below, give an expression in SQL for each of the queries that follows employee (ID, person_name, street, city) works (ID, company_name, salary) company (company_name, city) manages (ID, manager_id) (i) ✓ Find the employees whose name starts with 'C'. (ii) ✓ Find the name of managers of each company. (iii) ✓ Find the ID, name and city of residence of employees who works for "First Bank Corporation" and earns more than ₹50000. (iv) ✓ Find the name of companies whose employees earn a higher salary, on average, than the average salary at "First Bank Corporation".	8	L3	5	1,3,4
	(b) ✓ Let R = (A, B, C, D, E) be a relation scheme with the following dependencies AB → C, C → D, D → E, B → A, C → B Determine the total number of candidate keys and super keys.	4	L3	6	1,3,4
OR					
VII.	(a) Explain the concept of FD. Also state inference rules.	8	L2	6	1,3,4
	(b) Explain Triggers and Cursors with its syntax.	4	L2	5	1,3,4

(Continued)

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		Marks	BL	CO	PO
VIII.	(a) Describe shadow paging technique with a neat diagram.	5	L1	8	1,3
	(b) Discuss the timestamp ordering protocol for concurrency control. How does strict timestamp ordering differ from basic timestamp ordering?	7	L2	8	1,3
OR					
IX.	(a) ✓ What is the two-phase locking protocol? How does it guarantee serializability? Also discuss the variations of two-phase locking protocol.	7	L2	8	1,3
	(b) ✓ Why concurrency control is needed? Discuss any two concurrency problems with suitable examples.	5	L1	8	1,3

Blooms's Taxonomy Level's

L1 - 33.3%, L2 - 54.1%, L3 - 12.5%.
